

Chapter 5 — Exercises

Multiple-Choice Questions

Model evaluation & improvement

How to Use These Slides

- **12 multiple-choice** questions on Chapter 5 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: cross-validation, grid search, imbalanced data, confusion matrix, precision/recall/f1, thresholds, and ROC/AUC.

Q1. The main benefit of cross-validation over a single split is:

- A. a more stable, reliable estimate of generalization
- B. faster training
- C. fewer features
- D. guaranteed higher accuracy

Q2. Stratified k-fold cross-validation:

- A. shuffles the target labels
- B. keeps each class's proportion the same in every fold
- C. uses one sample per fold
- D. is only used for regression

Q3. Why shouldn't you tune parameters on the test set?

- A. It is too slow
- B. The test set is too small
- C. It leaks the test set and gives over-optimistic scores
- D. It changes the labels

Q4. `GridSearchCV.best_score_` reports:

- A. the test-set accuracy
- B. the training error
- C. the number of parameter combinations
- D. the mean cross-validation score on the training data

Q5. On a 99%-negative imbalanced dataset, accuracy is:

- A. misleading — "always predict negative" already scores 99%
- B. always zero
- C. the best possible metric
- D. equal to recall

Q6. Precision is defined as:

- A. $TP / (TP + FN)$
- B. $TP / (TP + FP)$
- C. $(TP + TN) / \text{all samples}$
- D. $FP / (FP + TN)$

Q7. Recall is high when the model:

- A. makes very few positive predictions
- B. has few false positives
- C. catches most of the actual positives
- D. predicts everything as negative

Q8. The f1-score is:

- A. the arithmetic mean of accuracy and recall
- B. the same as accuracy
- C. always higher than precision
- D. the harmonic mean of precision and recall

Q9. Lowering the decision threshold generally:

- A. increases recall but lowers precision
- B. increases both precision and recall
- C. has no effect
- D. increases precision but lowers recall

Q10. The AUC of a purely random classifier is:

- A. 1.0
- B. 0.5
- C. 0.0
- D. dependent on class balance

Q11. To make grid search optimize for AUC, set:

- A. `cv="roc_auc"`
- B. `metric="auc"`
- C. `scoring="roc_auc"`
- D. `average="auc"`

Q12. In a confusion matrix, the diagonal entries are:

- A. the false positives
- B. the false negatives
- C. all of the errors
- D. the correct predictions

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	A	CV averages several splits → a stable estimate
2	B	Stratified folds preserve class proportions
3	C	Tuning on the test set leaks it → optimistic scores
4	D	<code>best_score_</code> = mean CV score on training data
5	A	"Always negative" already scores 99% → accuracy misleads
6	B	Precision = TP / (TP + FP)

Answers 7–12

Q	Ans	Why
7	C	Recall = fraction of actual positives caught
8	D	f1 = harmonic mean of precision and recall
9	A	Lower threshold → more positives → ↑recall, ↓precision
10	B	Random classifier → AUC 0.5 , regardless of balance
11	C	Set <code>scoring="roc_auc"</code> in grid search / CV
12	D	Diagonal entries = correct predictions (TP, TN)

Well done!

Review Chapter 5 for any you missed